

Numerical Investigation of Window Design Strategies for Energy Efficiency and Thermal–Visual Comfort in Buildings

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ABSTRACT

Window design plays a decisive role in energy performance, daylighting, and occupants' comfort in buildings. This work investigates the effects of window shape and glazing type on energy consumption as well as indoor thermal and visual comfort in a five-story institutional building under the hot and humid climate of Bangladesh. The study compares twenty configurations, combining four window shapes (rectangular, arched, Palladian and bay) with five single-glazed tinted glass types (clear, blue, bronze, green, and grey; all 6 mm). Annual simulations were performed in DesignBuilder using the EnergyPlus engine to assess energy consumption, cooling demand, lighting load, solar heat gains, daylight availability, and comfort indicators such as Predicted Mean Vote (PMV), percentage of thermal, and visual comfort hours. Rectangular clear configuration was taken as the baseline. Results show that the Palladian green glass configuration offered the best performance, reducing annual energy use by 8.56% compared to the baseline, with a monthly average PMV of 1.88 and thermal comfort maintained for 32.1% of occupied hours. Bay clear windows showed the poorest performance with the highest energy use. Although the Palladian green configuration reduced cooling demand and increased thermal performance, the glass was darker and yielded a lower daylight, reducing visual comfort as compared to the clear glazing. In general, the results from this study suggest that the green glazing incorporated with the Palladian windows can contribute positively to building energy demand and thermal comfort in hot and humid climates, although complementary solutions could be needed to fulfill acceptable visual comfort.

Keywords: *Window Configuration, Window Glazing, Building Energy, Thermal Comfort, Visual Comfort.*